

Cloud Concepts I

Explore fundamental AWS cloud concepts such as global infrastructure components, migration advantages, security features, and responsibilities requiring root access. Understand the Well-Architected Framework pillar on sustainability and how AWS consulting partners can assist in cloud adoption.

Question 1

Which component of AWS global infrastructure is made up of discrete data centers and has redundant power and networking?

- A. Availability Zones
- B. Regions
- C. Points of Presence (PoPs)
- D. Outposts

Answer: A

Why: Availability Zones are data centers located in close proximity in a region with redundant power and network. Each region consists of at least two Availability Zones to take over in case one Availability Zone fails. They are designed for high availability and fault tolerance.

Why not:

Option B: A region consists of multiple Availability Zones with redundant power.

Option C: Points of Presence (PoPs), also known as edge locations, are in major cities. They cache content, reducing latency and improving end user performance.

Option D: AWS Outposts is a fully managed service that extends AWS services, tools, and APIs to on-premises customers.

Question 2

What are the advantages of migrating on-premises infrastructure to the cloud? **(Select any two options.)**

- A. Reduced cost of IT staff members
- B. Ability to scale within minutes to meet increasing user demands
- C. Reduced risk to business continuity due to redundancy and backup
- D. Complete control over infrastructure
- E. Enhanced security and privacy

Answer: B and C

Why:

Option B and C: AWS global infrastructure and services are designed to provide redundancy in case of disaster. Additionally, AWS offers built-in features to scale resources horizontally and vertically in little to no time.

Why not:

Option A: You would still need IT members to administer and manage resources on the cloud.

Option D: The majority of AWS services are fully managed. You would have very limited access, if any, to the infrastructure hosting the workloads.

Option E: Since the data and resources are hosted on the cloud, it is as secure as on-premises infrastructure.

Question 3

A company has decided to migrate its operations to AWS but lacks cloud expertise. Which feature can help the company?

- A. AWS Trusted Advisor
- B. AWS Artifacts
- C. AWS Consulting Partners
- D. AWS Systems Manager

Answer: D

Why: Sustainability encourages you to understand the impact of your business and maximize resource utilization.

Why not:

Option A: Security focuses on utilizing AWS services to improve the security of your infrastructure.

Option B: Reliability focuses on the ability of a workload to perform consistently and regularly as intended.

Option C: It encourages the utilization of resources to meet the requirements of the system effectively.

Question 5

Which task would require AWS root user access? **(Select any two options.)**

- A. Launch an AWS Lambda function.
- B. Activate the IAM access to the AWS Billing and Cost Management console.
- C. Configure MFA delete on an S3 bucket.
- D. View Amazon CloudTrail logs.
- E. Enable the encryption of an S3 bucket.

Answer: B and C

Why:

Option B: Only the root user can access the AWS Billing and Account Management console. Therefore root user account is required to activate IAM access to the console.

Option C: Similarly, enabling MFA delete requires root users as it adds an extra layer of authentication for deleting an S3 bucket.

Why not:

Option A, D, and E: These are routine tasks that do not require root user privileges.

Cloud Concepts II

Explore the AWS shared responsibility model, learn to manage accounts using AWS Organizations, and understand the Cloud Adoption Framework perspectives. This lesson also guides you on connecting with AWS experts via AWS IQ and selecting appropriate support plans for cost-effective cloud operations.

Question 6

A company has migrated its infrastructure to AWS to reduce its operational overhead. Which responsibility still belongs to the company?

- A. Controlling access to the AWS resources through IAM roles
- B. Terminate EC2 instances managed by the Auto Scaling group (ASG).
- C. Managing the underlying infrastructure of EC2 instances.
- D. Protecting AWS data centers from malicious traffic.

Answer: A

Why: According to the AWS shared responsibility model, the customer manages access to its resources using IAM roles and policies.

Why not:

Option B: Auto Scaling automatically manages the scaling of EC2 instances, including spinning up new instances when traffic spikes and terminating redundant ones when traffic is low.

Option C: AWS manages the underlying physical infrastructure of EC2 instances, including the servers, storage, and networking.

Option D: AWS handles the protection of physical infrastructure and network-level security at data centers. They secure the physical facilities and the cloud infrastructure from malicious attacks and unauthorized access.

Question 7

Which service is used to manage accounts for an entire organization?

- A. AWS Identity and Access Management (IAM)
- B. AWS Systems Manager
- C. AWS Organizations
- D. AWS Trusted Advisor

Answer: C

Why: AWS Organizations allows companies to manage and organize multiple AWS accounts. It enables users to group accounts into organizational units, apply policies across accounts, and centrally manage billing.

Why not:

Option A: IAM is used to manage user permissions and access within an AWS account; it does not provide account management for multiple accounts or organizations.

Option B: AWS Systems Manager is primarily used for managing and automating administrative tasks, like patching or configuration management, within instances and resources, but it does not manage accounts

Option D: Trusted Advisor helps in optimizing AWS environments, providing recommendations on cost savings, security, performance, and fault tolerance.

Question 8

Which perspective of the Cloud Adoption Framework (AWS CAF) ensures that cloud investments accelerate digital transformation?

- A. People
- B. Governance
- C. Operations
- D. Business

Answer: D

Why: The "Business" perspective of the AWS CAF focuses on aligning cloud initiatives with business goals. It emphasizes the importance of understanding how cloud investments can drive innovation, improve business agility, and enhance overall performance.

Why not:

Option A: The "People" perspective focuses on the workforce and culture necessary for successful cloud adoption, emphasizing skills and change management.

Option B: The "Governance" deals with establishing frameworks and policies for managing cloud usage and ensuring compliance rather than directly driving transformation.

Option C: The "Operations" perspective emphasizes optimizing IT operations and processes to support cloud services, but it does not primarily focus on business transformation.

Question 9

Your company is looking for certified freelance AWS experts to assist with a cloud project. Which AWS service can help you quickly connect with qualified professionals?

- A. AWS Enterprise Support
- B. AWS Trusted Advisor
- C. AWS IQ
- D. AWS Support Plans

Answer: C

Why: AWS IQ connects businesses with certified AWS experts for on-demand project-based work. It allows companies to browse profiles of AWS consultants, view their credentials, and hire them for specific tasks.

Why not:

Option A: AWS Enterprise Support offers proactive planning and advisory services.

Option B: AWS Trusted Advisor is used to evaluate existing AWS infrastructure.

Option D: AWS Support plans to offer technical support, but does not specifically include access to freelance consultants.

Question 10

Your company's development team working in a non-production environment is unable to delete a test resource on AWS. Considering this issue is occurring in a non-critical, cost-sensitive testing scenario, which AWS Support plan should you choose to get technical assistance?

- A. AWS Developer Support
- B. AWS Enterprise Support
- C. AWS Business Support
- D. AWS Trusted Advisor

Answer: A

Why: AWS Developer Support is tailored to developers working in non-production environments. It provides support for basic issues, such as troubleshooting resource deletion.

Why not:

Option B: AWS Enterprise Support is for large enterprises with mission-critical workloads. It offers extensive support, including a Technical Account Manager and faster response times (e.g., 15-minute initial response for business-critical issues).

Option C: AWS Business Support is intended for production environments where services require a higher degree of reliability and faster response times.

Option D: AWS Trusted Advisor is a tool that provides best practice recommendations and optimization guidance. It is not a support plan and does not offer direct technical assistance for troubleshooting issues like resource deletion. Its primary purpose is advisory rather than solving immediate operational problems.

Cloud Concepts III

Explore essential AWS cloud concepts through practice questions and solutions that cover shared responsibility, scalability, service models, and design principles. Learn to identify AWS resources, understand security responsibilities, and grasp fundamental cloud properties to prepare for the AWS Certified Cloud Practitioner exam.

Question 11

Which service or resource can be used to learn more about AWS from AWS experts?

- A. AWS Documentation
- B. AWS Developer Support
- C. AWS Online Tech Talks
- D. AWS Consulting Partners

Answer: C

Why: AWS Online Tech Talks provide insights from AWS experts on various topics related to AWS services and best practices.

Why not:

Option A: AWS Documentation is a comprehensive resource for AWS services, but lacks live expert interaction.

Option B: AWS Developer Support is a support plan that offers technical guidance instead of educational content.

Option D: AWS Consulting Partners assist with implementation and strategy but are not specifically geared toward learning.

Question 12

According to the AWS shared responsibility model, who is responsible for patching and fixing flaws in the infrastructure?

- A. It is the sole responsibility of the customer.
- B. It is the sole responsibility of AWS.
- C. It is a shared responsibility between AWS and the customer.
- D. The customer's AWS Support plan tier determines who manages the configuration.

Answer: C

Why: AWS is responsible for patching and fixing infrastructure layers; meanwhile, customers are responsible for patching their guest OS and applications.

Question 13

A company is migrating its legacy web application system to AWS. The application is rapidly growing, with a spike in traffic at certain hours of the day. Which cloud property will benefit the company the most?

- A. Scalability
- B. Flexibility
- C. Reliability
- D. Security

Answer: A

Why: Scalability is a fundamental property of cloud computing that allows resources to be adjusted based on demand. Since the company experiences spikes at certain hours of the day, it can save on costs by efficiently utilizing scaling services.

Why not:

Option B: Flexibility focuses on integrating multiple types of technologies together.

Option C: Reliability focuses on maintaining uptime and performance.

Option D: Security focuses on protecting data and resources.

Question 14

Why do AWS design principles encourage decoupled microservices architectures? **(Select any two options.)**

- A. It improves the scalability of the architecture.
- B. It improves the security of the architecture.
- C. It decreases the cost of the architecture.

- D. It allows you to update components regularly.
- E. It allows you to anticipate failure in the architecture.

Answers: A and D

Why: Decoupled microservices allow individual components of an application to scale independently based on demand. Also, a decoupled architecture enables teams to deploy updates to individual microservices without requiring a full system downtime.

Why not:

Option B: While microservices can lead to better isolation of security issues, security improvements are not a direct design principle of decoupling.

Option C: Decoupled architectures often introduce additional overhead in managing multiple microservices, communication layers, and monitoring.

Option E: Decoupled architectures are designed to tolerate failure, not necessarily to anticipate it. For example, if one microservice fails, others can continue functioning.

Question 15

Which cloud service model does AWS Lambda fall under, where you only need to manage your code without managing the underlying infrastructure?

- A. Infrastructure as a Service (IaaS)
- B. Platform as a Service (PaaS)
- C. Software as a Service (SaaS)
- D. Function as a Service (FaaS)

Answer: D

Why: AWS Lambda operates under the Function as a Service (FaaS) model, which allows users to run code in response to events without provisioning or managing servers.

Why not:

Option A: IaaS provides virtualized computing resources like servers, storage, and networks.

Option B: PaaS provides a platform for building and deploying applications without managing the underlying hardware.

Option C: SaaS provides software applications delivered over the internet.

Security and Compliance I

Explore essential AWS security and compliance concepts crucial for the AWS Certified Cloud Practitioner exam. Understand how to control network access, manage temporary credentials, protect sensitive data, ensure data integrity, and audit user actions using AWS services.

Question 16

Which AWS service would you use to restrict inbound traffic to an EC2 instance on port 5632, ensuring that only traffic on this specific port is allowed?

- A. Network Access Control Lists (NACLs)
- B. Elastic Network Interface (ENI)
- C. Security group
- D. AWS Web Application Firewall (WAF)

Answer: C

Why: Security groups act as firewalls for EC2 instances. This allows for restricting inbound and outbound traffic at the instance level.

Why not:

Option A: NACLs are used to restrict inbound and outbound traffic at the subnet level.

Option B: ENI is a logical networking component. It can not restrict access at the port level.

Option D: AWS WAF can only protect web applications by filtering HTTP and HTTPS traffic. WAF is primarily for application-level security, not network or port-specific access.

Question 17

What can be used to grant temporary access to AWS resources without using long-term credentials?

- A. IAM roles
- B. Access control lists (ACLs)
- C. Service control policies (SCPs)
- D. AWS Security Hub

Answer: A

Why: IAM roles have a policy attached to them, which grants temporary access to AWS users, accounts, and services.

Why not:

Option B: ACLs control access to resources like Amazon S3 buckets or VPC subnets, but they do not provide temporary access. They are more static and usually manage access control at the resource level. They don't handle temporary access.

Option C: SCPs are used to manage permissions across AWS Organizations. They apply permission boundaries to accounts but do not provide temporary access to specific resources.

Option D: Security Hub is a monitoring and compliance tool, not an access control mechanism. It provides security findings and best practices, but not access delegation.

Question 18

A company wants to use advanced machine learning services to recognize Personally Identifiable Information (PII) in data stored in an S3 bucket. The job needs to be executed once every week. Which service can meet these requirements with the least operational overhead?

- A. Amazon Comprehend
- B. AWS Lambda
- C. AWS Glue
- D. Amazon Macie

Answer: D

Why: Amazon Macie specifically analyzes, discovers, and protects sensitive information using machine learning and pattern-matching techniques. We can use it for scheduled jobs to analyze data stored in an S3 bucket.

Why not:

Option A: Amazon Comprehend is primarily used for natural language processing (NLP) tasks, such as identifying sentiment, entities, and language in unstructured text. It can also remove and identify PII. However, it will require an S3 batch operation for each object in the bucket. Also, we will need an orchestration service to enable weekly scans.

Option B: AWS Lambda is a serverless compute service used to execute code. We can use it to write our customer code to identify and remove PII, but it would cause an operational overhead.

Option C: AWS Glue is used to execute serverless ETL pipelines.

Question 19

Which S3 feature helps prevent accidental deletion of objects?

- A. S3 Storage classes
- B. S3 Lifecycle Policies
- C. S3 Versioning
- D. S3 Replication

Answer: C

Why: When Versioning is enabled, even if an object is deleted, a delete marker is created, and the previous versions remain accessible.

Why not:

Option A: Storage classes are used for cost optimization and do not prevent accidental deletion.

Option B: S3 Lifecycle Policies manage object expiration but don't prevent deletion.

Option D: S3 Replication allows you to replicate objects across different regions or buckets, but it does not prevent the accidental deletion of objects in the source bucket.

Question 20

You need to determine which IAM user uploaded a specific object to an Amazon S3 bucket. Which AWS service can provide this information?

- A. Amazon CloudWatch
- B. AWS CloudTrail
- C. AWS Security Hub
- D. Amazon Inspector

Answer: B

Why: AWS CloudTrail logs and monitors API calls made on your account, including actions taken on S3 buckets. When a file is uploaded to an S3 bucket, CloudTrail records the event, including the user's identity or the service that made the request.

Why not:

Option A: Amazon CloudWatch is primarily used for monitoring and logging the AWS resources and applications. It may log application behavior, but doesn't log API calls like S3 uploads with identity.

Option C: AWS Security Hub is a security service used to view compliance status across AWS accounts. It does not directly track specific user actions, such as file uploads.

Option D: Amazon Inspector is a security assessment service for applications deployed on AWS.

Security and Compliance II

Explore essential AWS security and compliance services used to detect threats, monitor alarms, protect against exploits, and manage configurations. This lesson helps you understand how tools like GuardDuty, Security Hub, and Secrets Manager work together to maintain a secure cloud environment and support centralized monitoring across multiple AWS accounts.

Question 21

Which service is used to monitor the AWS environment for malicious activity?

- A. AWS Shield
- B. AWS Inspector
- C. AWS Macie
- D. AWS GuardDuty

Answer: D

Why: AWS GuardDuty continuously monitors the AWS environment for malicious or unauthorized behavior and uses machine learning, anomaly detection, and integrated threat intelligence to identify and prioritize potential threats.

Why not:

Option A: AWS Shield protects against Distributed Denial of Service (DDoS) attacks.

Option B: AWS Inspector helps to improve the security and compliance of applications deployed on AWS.

Option C: AWS Macie helps identify and protect sensitive data.

Question 22

A company is using AWS security tools and services. They need a dashboard to view the status of the services and alarms. Which service will meet the requirements?

- A. Amazon GuardDuty
- B. Amazon Macie
- C. AWS Shield
- D. AWS Security Hub

Answer: D

Why: AWS Security Hub is designed specifically to provide a consolidated view of security findings across various AWS security services. It aggregates, organizes, and prioritizes security alerts and findings from multiple AWS services and presents them on a centralized dashboard.

Why not:

Option A: GuardDuty is a threat detection service that monitors and analyzes account activity to identify potential security threats.

Option B: Macie is a data security service that uses machine learning to discover, classify, and protect sensitive data within S3.

Option C: AWS Shield provides protection against Distributed Denial of Service (DDoS) attacks.

Question 23

Which service supports AWS WAF to protect against security exploits? **(Select any two options.)**

A. Amazon CloudFront

B. Amazon Route 53

C. AWS Application Load Balancer (ALB)

D. AWS Lambda

E. AWS Shield

Answer: A and C

Why: Both CloudFront and Application Load Balancer allow us to deploy WAF to protect the target against malicious internet traffic.

Why not:

Option B: Route 53 is a DNS service, and WAF does not integrate with DNS. It is only used to protect against HTTP(s) traffic.

Option D: AWS Lambda does not directly integrate with WAF. However, we can deploy a Lambda function behind an ALB to protect it.

Option E: AWS Shield is itself a security service that is used to protect against DDoS attacks.

Question 24

A company wants a sophisticated solution to manage configurations and passwords for its critical business applications. Which service will most cost-effectively meet these requirements?

A. AWS Secrets Manager

B. AWS Systems Manager Parameter Store

C. Amazon S3

D. AWS KMS

Answer: B

Why: AWS Systems Manager Parameter Store is designed to store and manage system configurations and passwords. Additionally, it uses IAM for access control to sensitive information.

Why not:

Option A: AWS Secrets Manager can only store a username and a password. It does not support storing system configurations.

Option C: Amazon S3 is primarily a storage service. It does not have features to protect sensitive information management and lacks native features for securely managing and rotating secrets.

Option D: The Key Management Service (KMS) creates and controls encryption keys. It can not store system configurations and passwords.

Question 25

Your company has multiple accounts for each team. They want to share access to CloudWatch data, such as logs, metrics, and alarms, to monitor the status of resources from a central account.

Which service can help?

A. AWS Security Hub

B. Amazon CloudWatch

C. AWS Resource Access Manager (RAM)

D. AWS Organizations

Answer: C

Why: AWS Resource Access Manager (RAM) is designed to securely share resources across AWS accounts. By using AWS RAM, we can share CloudWatch resources with multiple accounts.

Why not:

Option A: AWS Security Hub aggregates security-related information but does not provide sharing capabilities for CloudWatch data like logs, metrics, and alarms.

Option B: Though AWS CloudWatch provides monitoring capabilities, it doesn't support sharing resources across accounts.

Option D: AWS Organizations helps manage multiple AWS accounts, but doesn't support directly sharing CloudWatch data.

Security and Compliance III

Understand how to protect AWS resources using services like AWS Firewall Manager and Network Firewall. Learn about identity management differences, encryption options for S3 data, and compliance tools such as AWS Security Hub. This lesson prepares you to address security and compliance scenarios in the AWS Certified Cloud Practitioner exam.

Question 26

Which service is used to protect VPC from network threats?

A. AWS Firewall Manager

- B. AWS Network Firewall
- C. VPC Flow Logs
- D. AWS Web Application Firewall (WAF)

Answer: B

Why: AWS Network Firewall is a managed firewall service specifically designed to protect Amazon VPCs from network threats. It provides protection at the network level, including stateful inspection, intrusion prevention, and detection capabilities.

Why not:

Option A: AWS Firewall Manager is a management tool for centrally configuring and managing firewall rules, including rules for AWS WAF and AWS Network Firewall across multiple accounts and resources.

Option C: VPC Flow Logs capture information about the IP traffic going to and from network interfaces in a VPC. While useful for monitoring and troubleshooting, Flow Logs do not actively protect against threats.

Option D: AWS WAF protects web applications from application-layer threats like SQL injection and cross-site scripting (XSS).

Question 27

A company is considering using AWS Firewall Manager to improve its overall security posture. In which ways can it help? **(Select any three options.)**

- A. They can view the Flow Logs in and out of their applications.
- B. They can configure AWS Shield across their resources to protect them from DDoS attacks.
- C. They can configure access control policies to restrict the traffic.
- D. They can ensure that security group configurations across multiple VPCs adhere to organizational standards.
- E. They can configure it to respond to security alerts and findings of Security Hub by automatically applying protective rules across accounts.

Answer: B, C, and D

Why:

Option B: AWS Firewall Manager can centrally deploy and manage AWS Shield Advanced protections across multiple accounts, ensuring cohesive DDoS protection.

Option C: AWS Firewall Manager allows the centralized configuration and enforcement of policies (for instance, AWS WAF rules or security group policies) to control access and restrict traffic.

Option D: One of the key benefits of AWS Firewall Manager is that it can enforce security group policies across your AWS environment, ensuring compliance with organizational standards.

Why not:

Option A: Viewing Flow Logs is a feature of services like VPC Flow Logs, not AWS Firewall Manager. Firewall Manager is used for managing and enforcing security policies.

Option E: While Security Hub aggregates security findings, AWS Firewall Manager does not directly respond to alerts from Security Hub by automatically applying rules. Its focus is on policy management and enforcement rather than automated alert responses.

Question 28

Which option best describes the difference between IAM Identity Center and AWS Organizations?

- A. IAM Identity Center provides a single sign-on (SSO) option, while AWS Organizations manages multiple AWS accounts with governance and billing.
- B. IAM Identity Center is used exclusively for managing EC2 instances, while AWS Organizations is for managing Lambda functions.
- C. IAM Identity Center allows users to create custom policies, while AWS Organizations is solely for setting up multi-factor authentication.
- D. IAM Identity Center is only for billing purposes, whereas AWS Organizations is for user management across AWS services.

Answer: A

Why

IAM Identity Center provides single sign-on across AWS accounts and applications. It simplifies user authentication and authorization by centralizing identity access.

AWS Organizations provides centralized governance, consolidated billing, and management of multiple AWS accounts. It allows organizations to set policies and manage accounts efficiently.

Why not:

Option B: IAM Identity Center is not used for managing specific compute services like EC2; it manages identity and access across AWS services. Similarly, AWS Organizations is not a tool for managing individual services like Lambda.

Option C: Custom policies are created in IAM, not IAM Identity Center, and multi-factor authentication (MFA) is a security feature applied through IAM. AWS Organizations is not about MFA but about managing multiple accounts.

Option D: Billing in AWS is managed through the AWS Billing and Cost Management console and consolidated through AWS Organizations. User management across services is performed via IAM and IAM Identity Center, not solely through AWS Organizations.

Question 29

Your company wants to encrypt critical data stored in an S3 bucket. However, they want to manage and provide their own encryption keys with each HTTP request to the storage service.

Which encryption method would you recommend?

- A. Server-side encryption with S3 (SSE-S3)
- B. Server-side encryption with KMS (SSE-KMS)
- C. Client-side encryption with AWS KMS managed keys (CSE-KMS)
- D. Server-side encryption with customer-provided keys (SSE-C)

Answer: D

Why: Server-side encryption with customer-provided keys (SSE-C) allows customers to provide their own encryption keys and gain complete control over the encryption keys used to encrypt and decrypt their data.

Why not:

Option A: Server-side encryption with S3 (SSE-S3) uses AWS-managed keys. AWS manages the encryption and decryption process entirely.

Option B: Server-side encryption with KMS (SSE-KMS) uses AWS Key Management Service (KMS) to manage encryption keys. While you can manage key policies and grant access, AWS still retains some control over the keys.

Option C: Client-side encryption with AWS KMS managed keys (CSE-KMS) encrypts the data on the client side before sending it to S3, but it still relies on AWS KMS for key management.

Question 30

A health care organization is required to comply with regulatory standards such as HIPAA, PCI-DSS, and SOC 2. The compliance team needs to conduct regular security assessments and generate detailed reports to ensure compliance with these standards.

Which service will meet the requirements?

- A. AWS Trusted Advisor
- B. AWS GuardDuty
- C. AWS Inspector
- D. AWS Security Hub

Answer: C

Why: AWS Inspector automates security assessments of applications deployed on Amazon EC2 and helps identify vulnerabilities or deviations from best practices. Additionally, it provides detailed security reports for compliance audits. It generates detailed reports that help organizations meet compliance requirements, making it the best fit for the scenario.

Why Not:

Option A: Trusted Advisor provides guidance according to AWS best practices but does not provide compliance reports as required by regulatory standards.

Option B: GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior.

Option D: AWS Security Hub aggregates, organizes, and prioritizes security alerts from across AWS services, but doesn't provide compliance reports.

Security and Compliance IV

Explore important security and compliance concepts for AWS Certified Cloud Practitioner exam success. Understand how to implement rate limiting, perform SSL/TLS offloading, protect data at rest and in transit, and manage permissions using AWS services like WAF, ACM, KMS, and IAM controls. This lesson reinforces best practices to secure AWS environments and manage access effectively.

Question 31

Which service can be used to implement rate limiting on an Application Load Balancer (ALB)?

- A. AWS WAF (Web Application Firewall)
- B. Amazon API Gateway
- C. AWS Shield
- D. AWS Lambda

Answer: A

Why:

AWS WAF (Web Application Firewall) can be used to implement rate limiting on an ALB by creating rate-based rules that limit the number of requests from a specific IP address over a specified time period.

Why not:

Option B: While Amazon API Gateway can implement rate limiting for APIs, API Gateway is not directly related to an Application Load Balancer (ALB). ALBs are primarily used for distributing incoming application traffic, and rate limiting on ALB is handled through AWS WAF, not API Gateway.

Option C: AWS Shield provides DDoS protection but does not offer rate limiting functionality. It protects applications against large-scale attacks, but rate limiting for controlling request volume should be done using AWS WAF.

Option D: AWS Lambda can be triggered to perform various tasks, but it is not directly involved in implementing rate limiting on ALBs. Instead, Lambda can be part of a custom solution (like in combination with API Gateway or other AWS services), but for ALB rate limiting, AWS WAF is the right service.

Question 32

A financial institution wants to perform SSL/TLS offload to reduce the computation burden on its servers. Which services can help with least operational overhead?

- A. AWS Lambda
- B. EC2 Instances
- C. Amazon Certificate Manager (ACM)
- D. AWS CloudHSM

Answer: D

Why: AWS CloudHSM is specifically designed to handle SSL/TLS offloading.

Why not:

Option A: AWS Lambda is not suitable due to its event-driven nature and less computational time.

Option B: We can configure the EC2 instance to handle the verification and authentication process. However, it would result in a lot of operational overhead.

Option C: Amazon Certificate Manager (ACM) can help manage SSL/TLS certificates. However, it does not support SSL/TLS offloading.

Question 33

You want to protect your data at rest and in transit. Which AWS service or feature will meet the requirements? **(Select any two options.)**

- A. Amazon GuardDuty
- B. AWS Key Management Service (KMS)
- C. AWS CloudHSM
- D. AWS IAM

- C. AWS Certificate Manager (ACM)
- D. Amazon Batch Operations
- E. AWS Lambda

Answer: B and C

Why:

Option B: AWS KMS is designed to create and manage encryption keys used to encrypt data at rest.

Option C: ACM provisions, manages, and deploys SSL/TLS certificates. These certificates secure data in transit by encrypting the connection between clients and your servers.

Why Not:

Option A: GuardDuty is a threat detection service that helps identify malicious activity and unauthorized behavior in your AWS environment.

Option D: Amazon Batch Operations is designed for running batch processing jobs on AWS.

Option E: AWS Lambda is a serverless compute service; while it can process data, it does not inherently offer encryption features for protecting data at rest or in transit.

Question 34

What is the customer's responsibility in maintaining the security of an AWS Lambda function?

- A. Ensuring the Lambda function's execution role has the correct permissions
- B. Ensuring the physical security of the Lambda server infrastructure
- C. Patching the Lambda runtime environment
- D. Managing the availability of Lambda function execution resources

Answer: A

Why: The customer is responsible for managing the Lambda execution role, which determines the permissions the Lambda function has to access other AWS services.

Why not:

Option B: AWS manages the physical security of the servers that run Lambda functions. Customers are not responsible for the physical security of the underlying hardware.

Option C: AWS manages the patching and updating of the Lambda runtime environments, ensuring they are up to date with security patches. The customer does not need to manage runtime patching.

Option D: AWS automatically manages the availability and scaling of Lambda functions. Lambda functions are automatically scaled based on incoming requests; customers don't need to manage execution resources directly.

Question 35

Your AI team migrated their computational workloads to Amazon GPU-based EC2 instances. The team lead has access to create IAM roles and assign them to the team according to their requirements. However, you want to prevent your team lead from accidentally granting permissions to launch EC2 instances with larger configurations.

Which step would you take?

- A. Create a permission boundary that restricts the ability to launch larger instances.
- B. Use IAM Policies to deny "ec2:RunInstances" specific instance types.
- C. Assign the team lead a role that only allows viewing IAM policies without the ability to modify them.
- D. Implement AWS Service Control Policies (SCPs) to restrict the launching of larger EC2 instances across the organization.
- E. Use AWS Organizations to limit the ability to manage IAM roles in the team lead's organizational unit (OU).

Answer: A

Why: Permission boundaries act as a guardrail for IAM roles. They can effectively limit what permissions are granted by IAM roles, preventing the team leads from allowing the launching of larger EC2 instances.

Why not:

Option B: The team lead can override these permissions. This approach may not enforce the desired limit uniformly, as mistakes could occur.

Option C: This would overly restrict the team lead and hinder their ability to perform their necessary tasks of creating and managing roles.

Option D: SCPs provide restrictions at the organizational level, but applying them broadly might block legitimate needs in other parts of the organization. This option lacks the granularity to target only roles created by the team lead.

Option E: AWS Organizations can manage policies at scale, but they don't provide the detailed control at the individual role level that permission boundaries offer.

Cloud Technology and Services I

Explore essential AWS cloud technologies to understand cost-effective compute solutions, programmatic resource deployment, network connectivity, data migration techniques, and automated content moderation for real-world cloud applications.

Question 36

A company wants to run its critical application on the cloud. The application runs for 8–10 minutes when invoked by changes in the data state. The company wants an operational and cost-efficient computing solution.

Which option meets the requirements?

- A. Amazon EC2 On-Demand Instances
- B. AWS Lambda
- C. Amazon EC2 Reserved Instances
- D. Amazon EC2 Spot Instances

Answer: B

Why: AWS Lambda is a serverless computing service with a maximum execution time of 15 minutes. It is ideally suited for short-lived, event-driven tasks and is cost-efficient since

you only pay for the compute time you consume.

Why not

Option A: While On-Demand EC2 Instances offer operational simplicity, they require infrastructure provisioning and management. They are typically more costly than serverless solutions for intermittent and short-lived workloads.

Option C: Reserved Instances are cost-effective for steady-state workloads that run continuously. However, they require a long-term commitment and do not suit sporadic, short-duration workloads.

Option D: Spot Instances offer lower pricing for spare EC2 capacity but are best for fault-tolerant and non-critical applications.

Question No 37

Which AWS feature is used to programmatically deploy and configure resources?

- A. AWS Management Console.
- B. AWS CodePipeline
- C. AWS Cloud9
- D. AWS software development kits (SDKs)

Answer: D

Why: Amazon supports SDKs in multiple languages, including Python, JavaScript, etc., to deploy and configure resources programmatically.

Why not:

Option A: AWS Management Console provides a user interface to deploy and configure AWS resources.

Option B: AWS CodePipeline is a CI/CD pipeline service that automates the building, testing, and deployment of resources every time the code is changed.

Option C: AWS Cloud9 provides a web-based IDE to write, debug, and execute code.

Question 38

Which service can be used to connect multiple VPCs together in a mesh topology with the least operational overhead?

- A. VPC peering
- B. AWS Transit Gateway
- C. AWS Direct Connect
- D. VPC endpoints

Answer: B

Why: AWS Transit Gateway acts as a central hub that simplifies the network architecture by connecting multiple VPCs (and even on-premises networks) through one gateway. This approach minimizes the number of individual connections required, thereby reducing operational overhead and simplifying management.

Why not:

Option A VPC peering can only establish a point-to-point connection between two VPCs. Therefore, we would require n-1 VPC peerings to create a mesh topology between n VPCs.

Option C: AWS Direct Connect provides a dedicated private connection from our remote or on-premises network to the VPC. It can not be used to connect multiple VPCs.

Option D: VPC endpoints are used to privately connect supported AWS services and a VPC. It does not support VPC to VPC connection.

Question 39

A company wants to migrate exploratory footage from a remote location without a reliable internet connection. Which service can help migrate the data?

- A. AWS Data Sync
- B. AWS Direct Connect
- C. AWS Snowball
- D. AWS Site-to-Site VPN

Answer: C

Why: AWS Snowball devices are specifically designed for large amounts of data in environments with limited or unreliable internet connectivity. It can be shipped to a remote location, filled with data, and then sent back to AWS for direct upload into S3 or other AWS storage services.

Why not:

Option A: DataSync is used for online data transfers but requires a stable internet connection, which is not suitable for environments with unreliable connectivity.

Option B: Direct Connect provides a dedicated network connection from on-premises to AWS, but it is not suitable for remote locations without reliable internet access.

Option D: AWS Site-to-Site VPN establishes a secure connection between on-premises networks and AWS, but it is not suitable for remote locations without reliable internet access.

Question 40

A social media platform needs to moderate thousands of images and videos uploaded by users daily to ensure they follow community guidelines. Which option can be used to identify inappropriate or harmful content?

- A. Amazon Sagemaker
- B. Amazon Rekognition
- C. Amazon Textract
- D. AWS Comprehend

Answer: B

Why: Amazon Rekognition is designed to analyze images and videos, identifying inappropriate or harmful content like explicit imagery, violence, and offensive symbols.

Why not:

Option A: Amazon Sagemaker is used to build, run, and deploy machine learning models. It is not specifically tailored for content moderation.

Option C: Amazon Textract is used to extract text from images and documents. It does not analyze visual content for inappropriate or harmful material.

Option D: AWS Comprehend is used for natural language processing (NLP) tasks such as sentiment analysis, not image or video content analysis.

Cloud Technology and Services II

Explore key AWS services and their practical uses by reviewing practice exam solutions focused on cloud concepts, security, data management, and cost optimization. Understand how to select the right AWS service for specific use cases and improve your readiness for the AWS Certified Cloud Practitioner exam.

Question 41

A biotechnology research institute wants to integrate vast amounts of biological data from various sources, such as medical journals, clinical trials, and databases, into a unified system to uncover relationships between entities.

Which service can meet these requirements?

- A. Use Amazon RDS to store the data. Use SQL to query the data for patterns.
- B. Use Amazon Neptune to create a knowledge graph.
- C. Use Amazon Redshift to create a large data warehouse.
- D. Use Amazon DynamoDB to store the data and query it using PartiQL.

Answer: B

Why: Amazon Neptune is a fully managed graph database service designed to store and query complex, highly connected data. Biological data often involves intricate relationships between various entities, such as genes, proteins, diseases, and drugs. It also supports graph query languages like SPARQL and Gremlin.

Why not:

Option A: Amazon RDS is a relational database service that is well-suited for structured data with defined schemas. However, it is not optimized for handling highly connected data like biological relationships.

Option C: Amazon Redshift is a powerful data warehouse service designed for running complex analytical queries on large datasets. However, it is more suited for data analytics and reporting on tabular data rather than handling complex, graph-based relationships.

Option D: Amazon DynamoDB is a key-value and document database, used for simple, fast lookups rather than complex queries on highly interconnected data.

Question 42

What is the use case of EC2 reserved instances? **(Select any two options.)**

- A. An event-triggered workload that executes for a small period of time
- B. A critical workload that executes for a longer period of time
- C. An unpredictable workload that can be interrupted
- D. A predictable, steady-state workload that requires cost savings over a long period of time

Answers: B and D

Why: EC2 Reserved Instances are designed for predictable, critical, and steady-state workloads, offering significant cost savings compared to On-Demand Instances.

Why not:

Option A: This use case is better suited for on-demand instances that are designed to handle workloads that do not require a long-term commitment.

Option C: EC2 spot instances are better suited for unpredictable workloads, as they allow you to take advantage of spare EC2 capacity.

Question 43

A streaming company has just migrated its operations to AWS. Users from all over the world will be able to view short reels. Which service can meet their requirements?

- A. S3 bucket with a CloudFront distribution
- B. Lambda function with an API Gateway
- C. EC2 instances with Application Load Balancer
- D. AWS Global Accelerator

Answer: A

Why: The S3 bucket can be used as a scalable storage for short reels. We can configure a CloudFront distribution with an S3 bucket as the origin to cache and deliver the short reels.

Why not:

Option B: API Gateway and Lambda can not use content delivery optimization.

Option C: EC2 instances are more suited for handling dynamic content. This combination can not handle content delivery optimization and is costly.

Option D: The AWS Global Accelerator optimizes network traffic and is better suited to establishing quick UDP/TCP connections. It does not provide storage or caching functionality for content streaming.

Question 44

A company wants to host its warehouse on AWS, but it has an ML model used for data forecasting. It wants a solution that would enable it to run the custom model on data in the data warehouse. Which service meets the requirement?

- A. SageMaker
- B. Amazon Forecast
- C. Redshift ML
- D. Amazon Neptune

Answer: C

Why: Amazon Redshift ML allows SQL commands to run machine learning (ML) models directly on the data stored in Amazon Redshift. It supports the integration of custom ML models and uses SageMaker to automatically build and deploy the models within Redshift.

Why not:

Option A: Amazon SageMaker is used for building, training, and deploying ML models. However, it does not inherently integrate with data warehouses like Redshift.

Option B: Amazon Forecast is designed for time series forecasting. However, as described in the scenario, it doesn't support running custom models or in-warehouse processing.

Option D: Amazon Neptune is a graph database service, not designed for data warehousing or ML model execution.

Question 45

A company wants a database to handle IoT data streams. Which database will meet the requirements?

- A. Amazon RDS
- B. Aurora
- C. Amazon DocumentDB
- D. Amazon DynamoDB

Answer: D

Why: DynamoDB is a fully managed NoSQL database designed to handle high-velocity data and can automatically scale to accommodate large volumes of data from IoT devices. Since IoT streams are unstructured, DynamoDB's schema-less design is ideal for them.

Why not:

Options A and B: Amazon RDS and Aurora are relational databases that are not suitable for unstructured IoT streams.

Option C: Amazon DocumentDB is optimized for JSON-like document storage but may not be as effective in handling real-time streaming.

Question 46

A development team wants to automatically update their application whenever the source code changes. Which AWS service can help them deploy these changes to running instances or containers?

- A. AWS CodeCommit
- B. AWS CodeDeploy
- C. CloudFormation
- D. Amazon Elastic Container Service (Amazon ECS)

Answer: B

Why: AWS CodeDeploy is a fully managed deployment service that automates application deployment to various compute services, such as Amazon EC2, AWS Lambda, and on-premises servers. It updates running applications when the source code changes.

Why not:

Option A: AWS CodeCommit is a fully managed code repository that allows teams to collaborate on code. However, it cannot create CI/CD pipelines.

Option C: CloudFormation can launch resources and set up environments. However, it is not used for application code deployment in response to source code changes. It doesn't monitor source code or trigger deployments when code changes.

Option D: ECS is a container orchestration service that can run and manage Docker containers. However, it does not directly handle the update process upon code changes.

Question 47

A company wants to ingest real-time surveillance footage from multiple remote locations and process it for threat detection.

Which AWS service is best suited to efficiently ingest and process this streaming data?

- A. Amazon S3 Glacier
- B. AWS Snowball
- C. Amazon Kinesis
- D. Amazon Rekognition

Answer: C

Why: Amazon Kinesis allows real-time data ingestion and processing. It can handle high volumes of streaming data with low latency and integrate with other AWS analytics and machine learning services to analyze data and detect threats in near real-time.

Why not:

Option A: Amazon S3 Glacier is used for long-term data retention.

Option B: AWS Snowball is a physical data transfer device, not suitable for real-time video streaming.

Option D: Amazon Rekognition can be used for analyzing images and videos, but it's not designed for data ingestion. Rekognition would require the footage to be ingested first—typically via a service like Kinesis Video Streams—before it can analyze the content for threat detection.

Question 48

A company wants to launch a virtual machine to host a web application. Which service can help with least operational overhead?

- A. AWS ECS
- B. AWS Lightsail
- C. AWS EC2 Instances
- D. AWS Lambda

Answer: B

Why: Lightsail is designed for developers who must deploy and manage applications quickly and easily. It provides a simplified interface for getting started and includes compute power, storage, and networking.

Why not:

Option A: ECS requires management of the underlying infrastructure and orchestration, which can increase operational overhead.

Option C: EC2 requires configuration and management compared to Lightsail, leading to higher operational overhead.

Option D: Lambda is a serverless compute service that cannot be used to run a virtual machine.

Amazon Lambda is a serverless compute service that enables you to run a virtual machine.

Question 49

Which components provide outbound internet access to resources in the private subnet?

- A. Internet gateway
- B. Route table
- C. NAT gateway
- D. VPC endpoint

Answer: C

Why:

A NAT (Network Address Translation) gateway is specifically designed to allow resources in a private subnet to initiate outbound traffic to the internet (for example, to download updates or access external APIs) while preventing inbound internet access. The NAT gateway translates the private IP addresses to a public IP address when accessing the internet.

Why not:

Option A: An internet gateway is used to enable internet access for instances in public subnets (which typically have public IP addresses). However, resources in a private subnet do not have direct access via an internet Gateway.

Option B: A route table is a component used to define routes for traffic. While it is essential for directing traffic, it does not provide the network address translation functionality needed for outbound internet access from a private subnet.

Option D: A VPC endpoint provides a private connection between your VPC and supported AWS services (such as S3 or DynamoDB) without using the public internet. It is not designed to provide general outbound internet connectivity.

Question 50

A company is designing a mobile application. It needs authentication functionality that allows users to log in using social identity providers such as Gmail, Facebook, etc.

Which service can help it design the authentication system?

- A. Amazon Cognito
- B. Amazon Identity and Access Management
- C. Amazon Security Hub
- D. AWS Lambda

Answer: A

Why: Amazon Cognito handles web and mobile application authentication, authorization, and user management. It supports authentication through username and password or a third-party service like Facebook, Amazon, Google, or Apple.

Why not:

Option B: IAM is designed primarily for controlling access to AWS resources by managing users, groups, and roles, but it is not intended for authenticating end-users of mobile or web applications.

Option C: AWS Security Hub is used to manage and improve the security posture of your AWS resources. It does not provide user authentication capabilities.

Option D: AWS Lambda is a serverless compute service used to run code in response to events. It does not provide a user authentication service on its own.

Cloud Technology and Services IV

Explore how to analyze and troubleshoot network traffic with VPC Flow Logs, improve latency with AWS Local Zones, and deploy applications efficiently using Elastic Beanstalk. Understand storage options like EFS and FSx for high-performance computing on EC2 instances, and learn about secure VPC endpoints for private connections to AWS services.

Question 51

A company has deployed several EC2 instances in a private subnet for internal use. Recently, they have noticed some performance degradation and suspect unusual network traffic between these instances and external services. They want to analyze the traffic patterns to troubleshoot the issue.

Which AWS service can be used to collect detailed information about the traffic in a VPC?

- A. Amazon CloudWatch Logs
- B. Amazon CloudTrail
- C. VPC Flow Logs
- D. AWS Shield

Answer: C

Why: VPC Flow Logs allow you to capture IP traffic going to and from network interfaces in your VPC. Therefore, we can use them to monitor the traffic pattern of the VPC containing the EC2 instances.

Why not:

Option A: CloudWatch logs can be used to store logs from various AWS services. However, it does not collect detailed network traffic information. They serve as a destination for logs, not the source for collecting traffic data.

Option B: CloudTrail is focused on logging API calls and user activities within AWS accounts. It does not capture details about network traffic between EC2 instances and external services.

Option D: It is used to safeguard applications from distributed denial-of-service attacks but does not provide the detailed network traffic analysis required to troubleshoot performance degradation.

Question 52

A media streaming company wants to minimize latency and improve user experience for customers in a major metropolitan area. The company's primary AWS Region is too far away to provide optimal performance.

Which AWS service can provide local access to compute and storage resources with minimal operational overhead?

- A. AWS Local Zones
- B. AWS Outposts
- C. AWS Snowball
- D. AWS CloudFront

Answer: A

Why: AWS Local Zones provide compute, storage, and other services at edge locations closer to the users. Therefore, it will enable us to deploy the application closer to the users.

Why not:

Option B: AWS Outposts extends AWS infrastructure to on-premises locations, but it is better suited for hybrid deployments in customer-owned data centers.

Option C: AWS Snowball is used for physical data transfer, not continuous remote media processing.

Option D: AWS CloudFront is a content delivery network (CDN) for caching and delivering media. It does not provide the compute infrastructure to run a media processing application.

Question 53

A development team wants to deploy a .NET application on AWS without managing the underlying infrastructure. Which service offers the least operational overhead for this use case?

- A. Elastic Container Service (ECS)
- B. EC2 Instance
- C. Elastic Kubernetes Service (EKS)
- D. Elastic Beanstalk

Answer: D

Why: Elastic Beanstalk is a fully managed service that abstracts infrastructure management, including provisioning, load balancing, and scaling. This allows developers to focus on the application code rather than managing infrastructure.

Why not:

Option A: ECS is a container orchestration service; while it can reduce some overhead, it still requires configuring tasks, services, networking, and monitoring.

Option B: EC2 Instance requires full management of the underlying server, including OS patching, scaling, and configuration, leading to significantly higher operational overhead than Elastic Beanstalk.

Option C: EKS is a managed Kubernetes service. While it offloads some management, we still need to manage clusters, pods, deployments, and networking.

Question 54

You are running multiple high-performance computing applications on EC2 instances that require parallel access to the file system. Which storage option will meet the requirements?

- A. Elastic Block Store (EBS) Volumes
- B. Elastic File System (EFS)
- C. FSx
- D. Amazon S3

Answer: C

Why: FSx offers fully managed, high-performance file systems designed for workloads like HPC that need fast, parallel access.

Option A: EBS offers block storage and not a file system.

Option B: EFS is a scalable file storage service that supports parallel access across instances. However, it is optimized for general-purpose workloads, not necessarily for the high-performance needs of HPC workloads.

Option D: Amazon S3 is a general-purpose object storage service and lacks the high-performance characteristics needed for this scenario.

Question 55

Which option best describes VPC endpoints in AWS?

- A. A physical device that connects your on-premises data center to AWS over a private network
- B. A virtual router that provides NAT services for instances in a private subnet
- C. A secure and private connection between your VPC and supported AWS services without requiring internet access
- D. An internet gateway that enables outbound internet access for instances in a public subnet

Answer: C

Why: VPC endpoints allow private connections from the VPC to supported AWS services, such as S3 and DynamoDB, without going through the internet, ensuring better security and lower latency.

Option A: This option defines AWS Direct Connect, which is used for hybrid cloud setups with dedicated physical links.

Option B: This option defines a NAT gateway, which allows outbound internet from private subnets.

Option D: This option defines an internet gateway, which connects the VPC to the public internet, not AWS services, privately.

Billing, Pricing, and Support I

Explore key AWS billing and pricing services like Cost Anomaly Detection, Trusted Advisor, and Pricing Calculator. Understand how to monitor expenditure patterns, share reserved instances, and accurately estimate storage costs to optimize your cloud financial management.

Question 56

A company wants to instantly detect and respond to unexpected changes in the expenditure patterns of its AWS accounts. Which service will meet these requirements?

- A. AWS Cost Explorer
- B. AWS Savings Plans
- C. AWS Cost Anomaly Detection
- D. AWS Pricing Calculator

Answer: C

Why:

The AWS Cost Anomaly Detection service uses machine learning to continuously analyze our AWS cost and usage patterns, detecting anomalies or significant deviations from the norm. Also, it uses SNS and email notifications to alert us about unusual or unexpected changes in our AWS spending.

Why not:

Option A: AWS Cost Explorer is used for cost analysis and visualization. It can generate reports to investigate the cost expenditure, but it can not generate alerts for a more proactive approach.

Option B: Savings plans offer a flexible pricing model that helps us save costs compared to on-demand pricing. However, they do not help us monitor our costs.

Option D: Pricing Calculator offers a web-based interface to calculate the expenditure of our infrastructure. It will not help us monitor the costs.

Question 57

Your company has decided to migrate its infrastructure to AWS. Which services will help you evaluate your structure for performance and resilience, optimize your costs, and recommend remediations to achieve operational excellence?

- A. AWS Cost Explorer
- B. AWS Trusted Advisor
- C. AWS Health
- D. AWS Security Hub

Answer: B

Why:

AWS Trusted Advisor evaluates the AWS environment and suggests remediation actions for cost optimization, performance improvement, and operational excellence. It aligns our architecture according to the best AWS practices.

Why not:

Option A: AWS Cost Explorer is used to analyze and visualize the expenditure patterns of the AWS accounts. It can not recommend cost optimization techniques.

Option B: AWS Health focuses on the operational status of AWS services affecting the account, providing alerts for outages and maintenance.

Option D: AWS Security HUB provides a centralized view of our account's security posture. However, it does not provide any recommendations to improve the security of our account.

Question 58

Which S3 storage class will help us save the costs on objects with variable access patterns?

- A. S3 Standard
- B. S3 Infrequent Access
- C. S3 Glacier
- D. S3 Intelligent Tiering

Answer: D

Why:

S3 Intelligent Tiering automatically rotates objects between frequent access, infrequent access, and deep archive according to the access patterns. Therefore, it is ideal for objects with variable access patterns.

Why not:

Option A: S3 Standard would cost us a regular amount for rarely accessed objects, adding the costs redundantly.

Option B: S3 Infrequent Access provides lower cost for data accessed infrequently, but we must predict access patterns. There's also a retrieval cost, and it's not ideal for variable access.

Option C: S3 Glacier would incur high retrieval costs for objects accessed frequently.

Question 59

An organization has two departments: sales and engineering. The engineering department has bought some reserved instances that are not currently being used. On the other hand, the sales department has been utilizing some on-demand instances to run workloads for a few months. The company wants to manage the accounts so that both departments can share the reserved instances.

Which service can help them meet the requirements?

- A. AWS Organizations
- B. AWS Savings Plans
- C. AWS Systems Manager
- D. AWS Cost Explorer

Answer: A

Why:

AWS Organizations allows us to manage and consolidate accounts across departments under a single umbrella and to govern resources across departments.

Why not:

Option B: AWS Savings plans are used to save on costs as compared to the on-demand costs.

Option C: AWS Systems Manager provides a unified interface to view resources across AWS regions and accounts. However, it can not be used to share reserved instances.

Option D: AWS Cost Explorer analyzes and visualizes the expenditure pattern of AWS accounts, but does not control the sharing of reserved instances or manage multiple accounts.

Question 60

Which service can help calculate the storage costs accurately?

- A. AWS Pricing Calculator

- A. AWS Pricing Calculator
- B. AWS Trusted Advisor
- C. AWS Billing Conductor
- D. AWS Budgets

Answer: A

Why: AWS Pricing Calculator offers a web-based console that uses the AWS pricing API to accurately calculate the costs of AWS resources.

Why not:

Option B: Trusted Advisor offers recommendations regarding cost optimization, performance improvements, and security, but it does not specifically calculate storage costs.

Option C: Billing Conductor is used for creating customized billing solutions for businesses.

Option D: AWS Budgets helps track spending against a budget and provides alerts when you exceed thresholds.

Billing, Pricing, and Support II

Understand AWS billing concepts, pricing options like Saving Plans, and tools such as AWS Cost Explorer and Compute Optimizer. Learn to analyze usage, create custom bills, and apply cost management strategies to optimize your AWS cloud expenses.

Question 61

An EC2 instance wants consistently millisecond access to documents stored in an S3 bucket. The bucket's availability across regions is not important. Which S3 storage class will meet these requirements in the most cost-effective way?

- A. S3 Standard
- B. S3 Standard Infrequent Access
- C. S3 Standard One Zone
- D. S3 Glacier Instant Retrieval

Answer: C

Why:

S3 Standard One Zone provides consistent millisecond access within the availability zone. If we configure the S3 bucket in the Availability Zone of an EC2 instance, the S3 bucket will have a much lower access latency.

Why not:

Option A: Although it provides millisecond access, it is designed for high durability and availability across multiple Availability Zones. Also, it is not cost-optimized if regional redundancy isn't required.

Option B: S3 Standard Infrequent Access is used for infrequently accessed data that spans multiple availability zones.

Option D: S3 Glacier Instant Retrieval is used for archival and does not provide millisecond access latency.

Question 62

A company wants to use Amazon EC2 instances and S3 buckets for the next three years. They want to save on costs in the long term.

Which pricing models would you recommend?

- A. Pay-as-you-go approach
- B. AWS Budgets
- C. AWS Free Tier
- D. Saving plans

Answer: D

Why: Savings plans provide flexible pricing models that offer significant discounts for committed usage for one or three years, helping reduce costs for AWS services.

Why not:

Option A: Pay-as-you-go is used for short-term, on-demand needs and offers no long-term savings.

Option B: AWS Budgets is a billing and cost management service used to monitor and manage AWS costs.

Option C: AWS Free Tier is offered for new users. It offers limited services for up to 12 months.

Question 63

Which AWS service is used to create customized bills?

- A. AWS Cost Explorer
- B. AWS Pricing Calculator
- C. AWS Billing Conductor
- D. AWS Budgets

Answer: C

Why: AWS Billing Conductor allows users to customize their billing processes, including creating tailored billing rates and invoices for multiple departments after applying their respective savings plans.

Why not:

Option A: AWS Cost Explorer provides insights into spending patterns and resource usage, but it does not create customized bills.

Option B: AWS Pricing Calculator is a web-based console to calculate the estimated cost of AWS resources.

Option D: AWS Budgets is used to track and monitor the expenditure. It can generate alarms for exceeded costs, but can not create bills.

Question 64

A company has been monitoring and recording their workload utilization of EC2 instances for the past year. They want to use this data to improve their expenditure. Which service can help?

- A. AWS Systems Manager
- B. AWS Compute Optimizer

- B. AWS Compute Optimizer
- C. AWS Pricing Calculator
- D. AWS Budgets

Answer: B

Why: AWS Compute Optimizer analyzes the historical utilization of EC2 instances and provides recommendations to optimize costs. It helps identify underutilized or over-provisioned instances and suggests rightsizing options.

Why not:

Option A: AWS Systems Manager provides information about the AWS resources in the environment but can not identify underutilized or over-provisioned instances.

Option C: AWS Pricing Calculator is used to estimate the costs of AWS services.

Option D: AWS Budgets is used to set budgets and track and monitor them.

Question 65

Which tasks can be performed using AWS Cost and Usage Reports (CUR)?

- A. Analyze AWS usage by service and region in great detail.
- B. Set alerts based on cost thresholds for specific AWS resources.
- C. Forecast future AWS costs based on historical data.
- D. Receive personalized recommendations for cost optimization.

Answer: A

Why: AWS Cost and Usage Reports provide detailed billing and usage information, including service usage by region.

Why not:

Option B: AWS Budgets allow us to set the expenditure budget and generate alarms when the cost exceeds.

Option C: AWS Cost Explorer provides features for forecasting costs.

Option D: AWS Trusted Advisor evaluates the architecture and can provide more personalized recommendations for cost optimization.